

THE ROBOTIC FARM

How Unmanned Ground Vehicles Are Transforming Modern Agriculture

Keyur Kulkarni

Introduction



Modern farming is undergoing a major technological shift, with output no longer dependent solely on manual labor and traditional equipment. Breakthroughs in robotics, artificial intelligence (AI), the Internet of Things (IoT), and self-guided navigation systems have paved the way for what is known as precision agriculture — a method that boosts yield while cutting down on wasted resources. Unmanned Ground Vehicles (UGVs) stand out as one of the most exciting developments driving this change.

What are Unmanned Ground Vehicles?

An Unmanned Ground Vehicle (UGV) is a ground-based robotic system designed to function with full or partial autonomy. In contrast to conventional tractors, which need continuous human input to operate, UGVs can navigate crop rows on their own, detect and steer clear of obstacles, gather data from the field, and carry out farming tasks through built-in decision-making capabilities.



Modern agricultural UGVs typically integrate:

- GPS and RTK positioning for centimetre-level navigation
 - LiDAR and ultrasonic sensors for obstacle detection
 - RGB, multispectral, and thermal cameras
 - AI-based computer vision systems
 - Robotic arms or spraying mechanisms
 - Wireless communication through IoT networks
 - Battery-electric or hybrid power systems
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How your sensor system would actually work :

Navigation

GPS / GNSS / RTK



Perception

LiDAR/cameras / ultrasonic



Intelligence

AI/computer vision/sensor fusion



Control

Embedded controller/navigation algorithm



Actuation

Sprayer/weeder / robotic arm



Communication

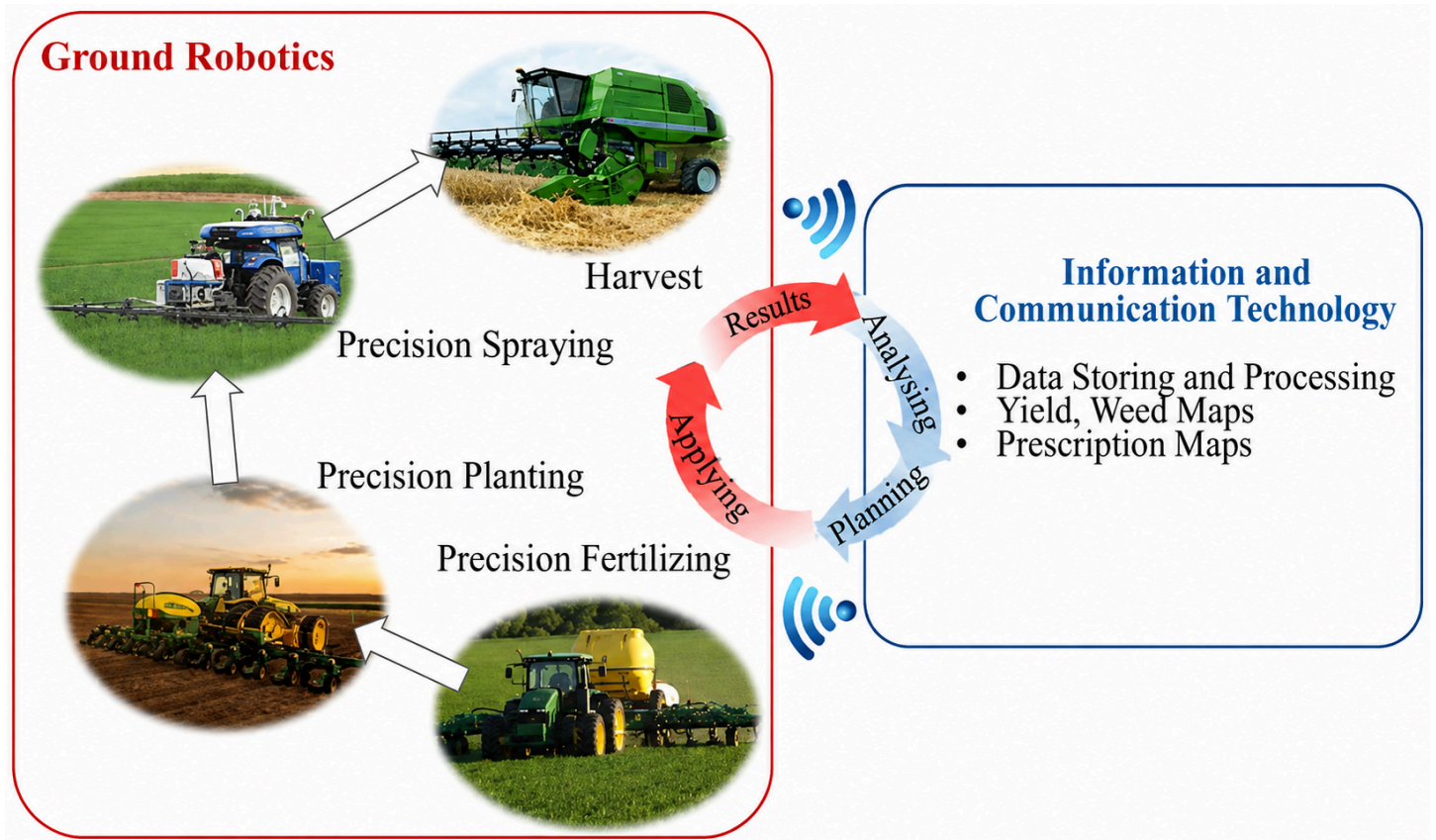
IoT / telemetry/farmer dashboard



Power

Battery / hybrid system

Applications of UGVs in Modern Agriculture



Precision Control

Weed

Automated weed management stands out as one of the most effective use cases for UGVs. This targeted method cuts down substantially on herbicide use, reduces overall production expenses, and limits harm to the environment. Studies indicate that AI-powered weed control systems in UGVs have evolved quickly, moving from conventional machine learning approaches toward deep learning methods that deliver higher precision in real-world field conditions.

Precision Spraying

Applying fertilisers, pesticides, and herbicides accurately. Minimising chemical waste and improving crop protection. Reducing farm workers' exposure to harmful chemicals.

Crop Monitoring

UGVs continuously inspect crops using high-resolution cameras and multispectral imaging systems.

Parameters they work on:

- Nutrient deficiencies
- Water stress
- Pest infestations
- Plant diseases
- Growth abnormalities

allowing farmers to intervene before problems spread, improving overall yield and reducing economic losses.

Disease Detection

Because UGVs operate close to the plants, they can detect early-stage infections before they spread, and by tagging affected locations with GPS, they enable targeted treatment rather than blanket application across the field. Many systems also process data in real time or near real time, allowing farmers to receive prompt alerts, while routine operation supports continuous monitoring of disease progression and treatment effectiveness over time.

Soil monitoring

UGVs support soil monitoring by carrying sensors that measure moisture, temperature, pH, nutrient levels, and compaction as they move through the field, offering precise, location specific data. Traveling directly across the land allows them to take readings at multiple points and depths

with greater accuracy than aerial methods. This geo-tagged data is used to create detailed soil maps, helping farmers apply water, fertilizer, or amendments only where needed, reducing waste and supporting better soil health and long-term land management.

Yield estimation

UGVs support yield harvesting by using sensors and AI-based image recognition to detect crop ripeness and identify the best time to harvest. Some are equipped with robotic arms that can autonomously pick produce with precision, reducing damage and labour dependence. As they move through fields, UGVs also estimate yield by counting and measuring crops, helping farmers plan storage, transport, and labour needs. This also enables selective harvesting of ready zones rather than the whole field at once, improving quality and reducing waste.

Seeding Planting

and

Carry out automated seed planting with precise spacing and depth control. Enhance planting efficiency and promote more uniform crop growth. Minimise seed loss.



Advantages

- Reduced Labour Requirement
- Higher Precision
- Increased Productivity
- Lower Input Costs

Disadvantages

- High Initial Cost
- Technical Complexity
- Limited Battery Life
- Navigation Challenges
- Weather Dependence

Where the future lies



Fully Autonomous Farming

The future of agricultural robotics is more likely to be progressive autonomy than immediate replacement of human operators.



AI-Based Decision Making

AI will enable UGVs to identify diseases, pests, weeds, and nutrient deficiencies in real time.



Swarm Robotics

Groups of small UGVs will cooperate to perform tasks such as seeding, fertilizing, and harvesting.



Smart Precision Agriculture

Farmers will receive real-time recommendations for irrigation, fertilization, and pest management.

A UGV's Day in an Indian Vegetable Farm



UGVs can transform Indian agriculture by automating labour-intensive tasks such as precision seeding, weeding, spraying, and crop monitoring—critical given the country's small landholdings and seasonal labour shortages. These machines improve accuracy, reduce input wastage and chemical exposure for farmers, and can operate continuously

regardless of weather. Compact, affordable UGV models could make precision farming accessible even to smallholders, helping offset the impact of rural labour migration. Wider adoption, however, depends on overcoming high costs, limited technical training, and gaps in rural infrastructure, such as connectivity and after-sales support.

A UGV's Day in an Indian Vegetable Farm

06:00 — Mapping

UGV maps the field using RTK-GNSS.



07:00 — Crop inspection

Camera identifies crop and weed locations.



08:00 — AI analysis

AI generates a weed-density map.



09:00 — Targeted treatment

UGV mechanically removes or selectively sprays weeds.



11:00 — Soil monitoring

Sensors identify areas requiring irrigation.



14:00 — Data upload

Farm data reaches farmer/FPO dashboard.



Next operation

The farmer decides what the robot should do next.

The Future Flowchart

Agriculture 1.0

Human/animal labour



Agriculture 2.0

Mechanisation



Agriculture 3.0

Electronics + precision agriculture



Agriculture 4.0

IoT + AI + connected farming



Agriculture 5.0

Human + AI + Robotics collaboration

The farm of tomorrow will not be human or autonomous—it will be collaborative.

Unmanned Ground Vehicles point toward a smarter, more connected way of farming, not just automated tasks. By combining sensors, AI, navigation, and robotics, they help farmers apply resources precisely where needed, cutting waste.

But the real promise lies in collaboration, not replacement. Farmers bring experience and judgment; UGVs offer constant monitoring and the stamina for repetitive, demanding work.

In India, success depends less on sophistication and more on affordability, modularity, and adaptability to small, fragmented farms. The challenge is turning advanced technology into tools farmers can actually use.

The question is no longer whether robots belong in agriculture—it's how well we integrate them, as partners to farmers rather than substitutes.



SOURCES

- ICAR – Robotics & AI in Agriculture, 2026
- ICAR – Agricultural Engineering & Technology
- Computers and Electronics in Agriculture, 2024
- Machines, 2024
- Journal of Sensor and Actuator Networks, 2024
- U.S. GAO – Precision Agriculture, 2024
- arXiv – Agricultural UGV Navigation Research
- arXiv – UAV–UGV Precision Farming Research